

Pattern Matching and Machine Learning for Audio Signal Processing

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Chapter 7: Discriminative classification via SVMs

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Overview

1. Motivation: classification problem
2. Linear classification (using SVMs)
3. Non-linear classification (using SVMs)
4. Final remarks

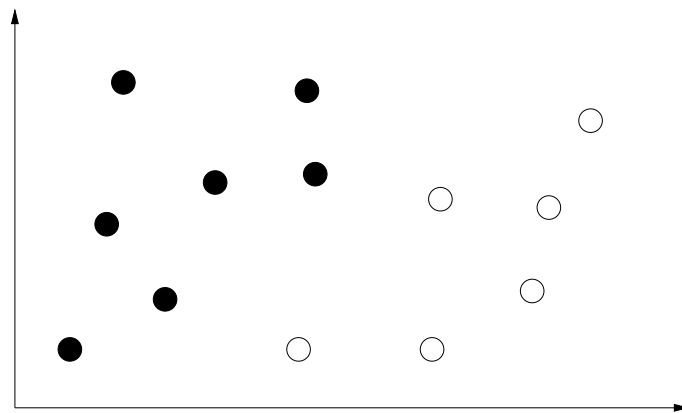
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Chapter 7: Discriminative classification via SVMs (Short version) – 1 / 31

1. Motivation: classification problem

Given: Data $(x_1, y_1), \dots, (x_n, y_n) \in \mathbb{R}^N \times \{-1, 1\}$

Example: $N = 2, n = 13$, visualization: $-1 \equiv \bullet, 1 \equiv \circ$



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Chapter 7: Discriminative classification via SVMs (Short version) – 2 / 31

1. Motivation: classification problem

Interpretation: Data (x_i, y_i) are *samples* of a binary classification function

$$f : \mathbb{R}^N \rightarrow \{-1, 1\},$$

that assigns class labels 1 or -1 to each vector $x_i \in \mathbb{R}^N$.

Task: Determine f , given the above n samples

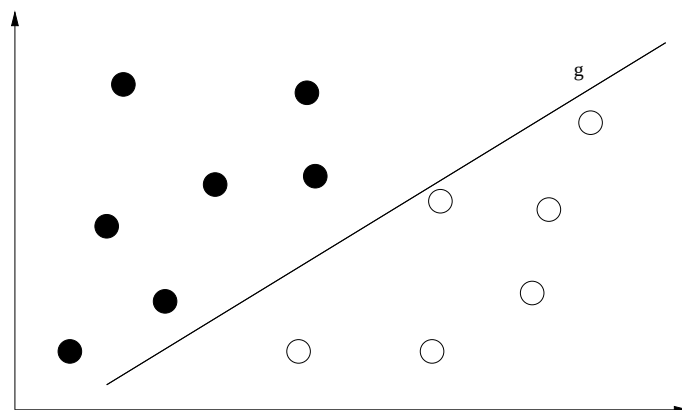
(Learning problem: “Learn f from training data”, Approximation problem: “Determine f from sampling points”)

Warning: The above task has to be further specified (too many f ’s possible) ...

→ Have to restrict space of possible functions!

2. Linear classification (using SVMs)

Example: (Continued) Straight line $g \rightsquigarrow$ implies two classes



Classification using such a linear separation function is called
linear classification

Some prerequisites from linear algebra

Representation of a $N - 1$ dimensional hyperplane g_w (through the origin) by normal vector $w \in \mathbb{R}^N$:

$$g_w := \{x \mid \langle w, x \rangle = 0\}$$

This extends to a hyperplane through a point $b' \in \mathbb{R}^N$:

$$\begin{aligned} g_w + b' &= \{x + b' \mid \langle w, x \rangle = 0\} \\ &= \{y \mid \langle w, y \rangle - \langle w, b' \rangle = 0\} \\ &=: \{y \mid \langle w, y \rangle - b = 0\} \quad \text{for } b \in \mathbb{R} \\ &=: g_{w,b}. \end{aligned}$$

Classification of a point $y \in \mathbb{R}^N$

Let $y \notin g_{w,b}$. Write $y =: y' + b'$. Then

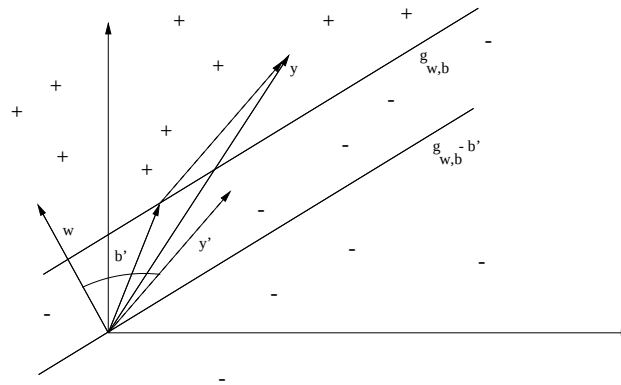
$$\begin{aligned} \langle w, y \rangle - b &= \langle w, y' \rangle \\ &= \|w\| \|y'\| \cos(w, y') \quad \text{with } \|w\| \|y'\| > 0 \\ &\begin{cases} > 0 & \text{if } y' \text{ is located "above" } g_{w,b} - b' \\ < 0 & \text{if } y' \text{ is located "below" } g_{w,b} - b' \end{cases} \\ &\begin{cases} > 0 & \text{if } y \text{ is located "above" } g_{w,b} \\ < 0 & \text{if } y \text{ is located "below" } g_{w,b} \end{cases} \end{aligned}$$

$\rightsquigarrow$ choose classification function

$$f_{w,b} : x \mapsto \text{sign}(\langle w, x \rangle - b)$$

Classification of a point $y \in \mathbb{R}^N$

Illustration:



Classification function

$$f_{w,b} : x \mapsto \text{sign}(\langle w, x \rangle - b)$$

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Remark

Let $f_{w,b}(x) > 0$, then $\langle w, x \rangle - b = \delta > 0$ and hence

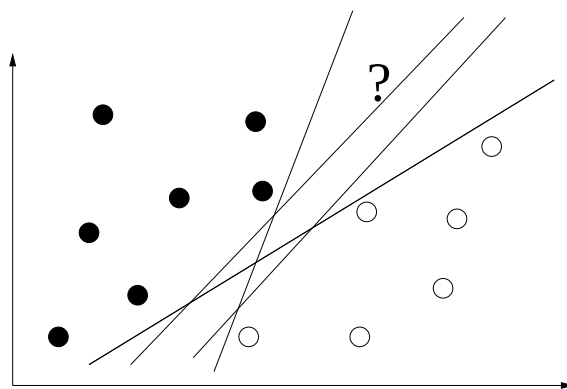
$$\left\langle \frac{w}{\delta}, x \right\rangle - \frac{b}{\delta} = 1.$$

We shall subsequently use this “normal form” ...

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New problem: How to choose $f_{w,b}(x)$, given some labeled data?

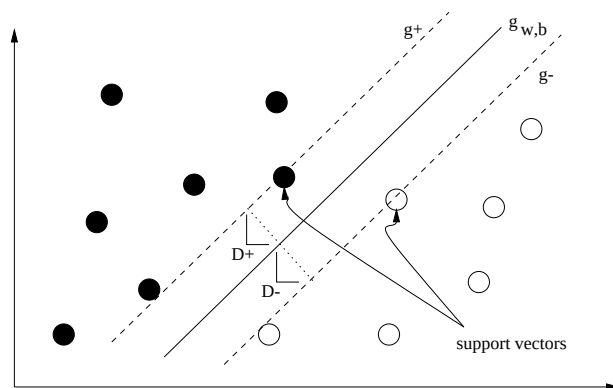


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Support vector approach

Choose w, b , such that the minimum distance between $g_{w,b}$ and $\{x_1, \dots, x_N\}$ is maximized:



What is $\Delta := D_+ + D_-$ for given w, b ?

(Remark: obviously $D_+ = D_-$.)

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Computation of Δ

Let $x_i \in g_+$ then w.l.o.g.

$$\langle w, x_i \rangle - b = 1.$$

(Points on the lines g_+ and g_- are called **support vectors**.)

Furthermore, there is a support vector $x_j \in g_-$ and

$$\langle w, x_j \rangle - b = -1.$$

Then it holds that

$$\Delta = \left\langle \frac{w}{\|w\|}, x_i - x_j \right\rangle = \frac{1}{\|w\|} (b + 1 - (b - 1)) = \frac{2}{\|w\|}.$$

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Goal: Maximize Δ

Alternatively: **Minimize** $\|w\|_2$ (equivalently $\|w\|_2^2$)
provided that all samples are correctly classified.

To formalize this, we note that

$$\forall i \in [1 : n] : |\langle w, x_i \rangle - b| \geq 1.$$

It follows that we have to solve the following optimization problem:

Minimize $\frac{1}{2} \|w\|_2^2$ provided that

$$\forall i \in [1 : n] : y_i \cdot (\langle w, x_i \rangle - b) \geq 1$$

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Optimization using Lagrange multipliers

Theorem: Let $B \subset \mathbb{R}^n$ an open set and let $f : B \rightarrow \mathbb{R}$ be continuously differentiable. Let furthermore $\varphi_1, \dots, \varphi_k : B \rightarrow \mathbb{R}$ be continuously differentiable such that

$$\text{rank} \left(\frac{\partial \varphi_i}{\partial x_j}(x) \right)_{i,j} = k < n \text{ on } B,$$

Then the following holds:

If f has a local extremal value $p \in B$ such that $\varphi_1(p) = \dots = \varphi_k(p) = 0$, then there are uniquely determined $\alpha_1, \dots, \alpha_k \in \mathbb{R}$ (**Lagrange multipliers**), such that $\forall i \in [1 : k]$:

$$\frac{\partial f}{\partial x_i} = \sum_{j=1}^k \alpha_j \frac{\partial \varphi_j}{\partial x_i}.$$

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Application of Lagrange multipliers

Solve $n + k$ equations $\varphi_i(p) = 0, i \in [1 : k]$ and

$$\frac{\partial}{\partial x_j} \left(f - \sum_{i=1}^k \alpha_i \varphi_i \right) (p) = 0, \quad j \in [1 : n].$$

To apply this to our above problem, define $\varphi_i = \varphi_i(w, b) := y_i(\langle w, x_i \rangle + b) - 1$ and

$$L(w, b, \alpha) := \frac{1}{2} \|w\|^2 - \sum_{i=1}^n \alpha_i [y_i(\langle w, x_i \rangle + b) - 1]. \quad (*)$$

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Elimination of w and b (required later on!)

We consider the following necessary conditions for extremal values:

$$\frac{\partial L}{\partial b} = 0 \quad \text{und} \quad \frac{\partial L}{\partial w} = 0.$$

With those we obtain:

$$-\sum_{i=1}^n \alpha_i y_i = 0$$

and

$$0 = w - \sum_{i=1}^n \alpha_i y_i x_i \Leftrightarrow w = \sum_{i=1}^n \alpha_i y_i x_i.$$

Substitution into (*)

yields

$$\begin{aligned} & \frac{1}{2} \|w\|^2 - \sum_{i=1}^n \alpha_i [y_i (\langle w, x_i \rangle + b) - 1] \\ &= \frac{1}{2} \left\langle \sum_{i=1}^n \alpha_i y_i x_i, \sum_{i=1}^n \alpha_i y_i x_i \right\rangle - \sum_{i=1}^n \alpha_i \left[y_i \left(\left\langle \sum_{j=1}^n \alpha_j y_j x_j, x_i \right\rangle + b \right) - 1 \right] \\ &= \dots \\ &= \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle \end{aligned}$$

...and hence the “dual optimization problem”

where we only have to optimize w.r.t. α :

Maximize

$$F(\alpha) := \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle$$

with constraints

$$\sum_{i=1}^n \alpha_i y_i = 0.$$

Optimization using Kuhn-Tucker Multipliers

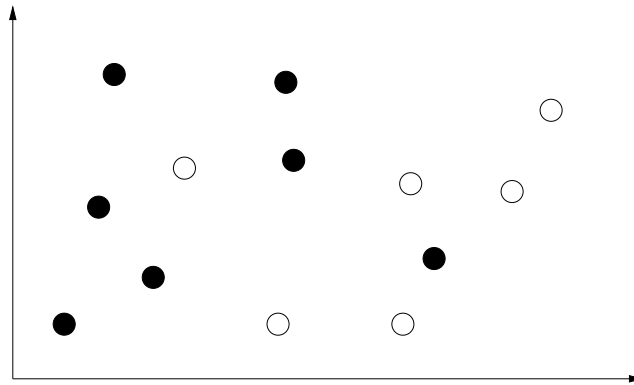
Problem: The above constraints include **inequalities** — The use of Lagrange multipliers, however, requires **equalities**.

Solution: We will use so called **Kuhn-Tucker multipliers** — in our case this essentially leads to additional constraints

$$\forall i : \alpha_i \geq 0$$

(Remark: The resulting quadratic optimization problem can be efficiently solved using quadratic programming (QP).)

Problem: Usually, real data are not linearly separable

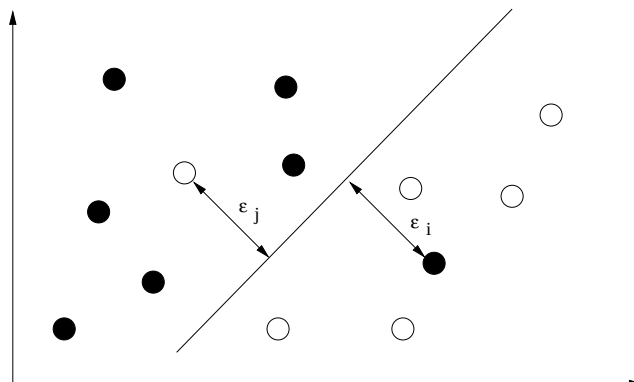


(Def. of *not linearly separable*: ... there is no straight line $g_{w,b}$ such that the above constraints are satisfied.)

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Solution: Use of slack variables



Modify constraints as follows: $\forall i \in [1 : n]$

$$y_i \cdot (\langle w, x_i \rangle + b) \geq 1 - \varepsilon_i \quad \text{for an } \varepsilon \geq 0.$$

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New optimization problem

Instead of $\frac{1}{2}\|w\|^2$, now we optimize $\frac{1}{2}\|w\|^2 + C \sum_{i=1}^n \varepsilon_i$ w.r.t w, b, ε and a **regularization parameter** C . Analogously to the previous scenario this leads to the optimization problem

Maximize

$$F(\alpha) := \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y_i y_j \langle x_i, x_j \rangle$$

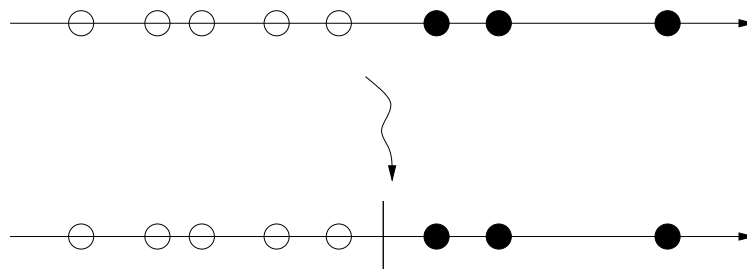
with constraints

$$\forall i \in [1 : n] : 0 \leq \alpha_i \leq C \quad \text{und} \quad \sum_{i=1}^n \alpha_i y_i = 0.$$

3. Non-linear classification (using SVMs)

We consider one-dimensional linear classification.

Example:



Non-linear classification

Problem: How to linearly separate



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Non-linear classification

Idea: Transform data $\{x_1, \dots, x_n\}$ to feature space by using a [feature extractor](#)

$$\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^M$$

(usually $M > N$).

... where data can be linearly separated!

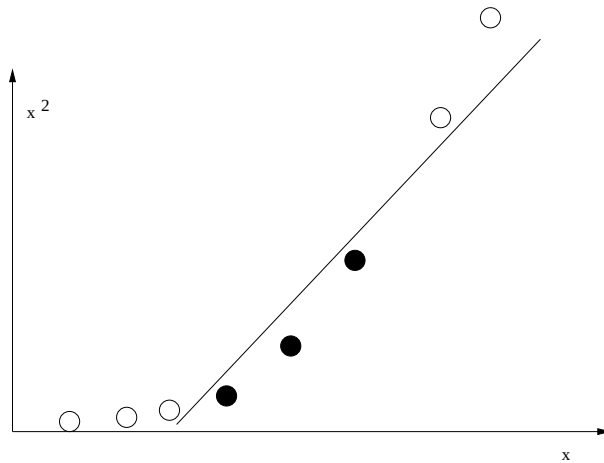
In our example: $\Phi : \mathbb{R} \rightarrow \mathbb{R}^2$

$$x \mapsto (x, x^2)$$

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Using Φ leads to



The data $\{(x_i, x_i^2, c_i) | i \in [1 : n]\}$, $c_i \in \{-1, 1\}$ are linearly separable after applying Φ !

The conditions for non-linear separability

... are given by

$$y_i \cdot (\langle w, \Phi(x_i) \rangle + b) \geq 1$$

$\forall i \in [1 : n]$. The **optimization problem** is modified to

$$F_{\Phi}(\alpha) := \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y_i y_j \langle \Phi(x_i), \Phi(x_j) \rangle$$

plus constraints.

Problem: “Good” choices of $\Phi : \mathbb{R}^N \rightarrow \mathbb{R}^M$ (e.g. all monomials of degree N) lead to high dimensional features.

↪ Evaluating the inner products in the feature space is too expensive!

Efficient evaluation using kernel functions

Kernel trick: For certain Φ 's, the inner product

$$\langle \Phi(\cdot), \Phi(\cdot) \rangle =: k(\cdot, \cdot)$$

can be evaluated efficiently!

Example: $\Phi(x_1, x_2) := (x_1^2, \sqrt{2}x_1x_2, x_2^2)$

Then

$$\begin{aligned} \langle \Phi(x), \Phi(y) \rangle &= \langle \Phi(x_1, x_2), \Phi(y_1, y_2) \rangle \\ &= (x_1^2, \sqrt{2}x_1x_2, x_2^2) \cdot (y_1^2, \sqrt{2}y_1y_2, y_2^2)^\top \\ &= ((x_1, x_2)(y_1, y_2)^\top)^2 =: k((x_1, x_2), (y_1, y_2)) = k(x, y). \end{aligned}$$

Generally it holds that $\langle \Phi(x), \Phi(y) \rangle = \langle x, y \rangle^m$ if Φ maps to all monomials of degree m .

Remark: $k(\cdot, \cdot)$ is called **kernel function**.

Training in the non-linear case

Substitution into the above target function yields

$$F_\Phi(\alpha) = \sum_{i=1}^n \alpha_i - \frac{1}{2} \sum_{i,j=1}^n \alpha_i \alpha_j y_i y_j k(x_i, x_j).$$

classification in the non-linear case

Analogously to the representation $w = \sum_{i=1}^n \alpha_i y_i x_i$ one observes that

$$w = \sum_{i=1}^n \alpha_i y_i \Phi(x_i)$$

and hence

$$\begin{aligned} \langle w, \Phi(x) \rangle &= \sum_{i=1}^n \alpha_i y_i \langle \Phi(x_i), \Phi(x) \rangle \\ &= \sum_{i=1}^n \alpha_i y_i k(x_i, x). \end{aligned}$$

Using this result we may construct a kernel-based classification function.

4. Final remarks

- SVMs started to become popular since 2001,
- they perform well for many classification problems.

Challenges:

- Binary classification only (using the classical concept).
- Find suitable feature extractors Φ 's (admitting corresponding kernels k 's).
- The latter is extensively discussed in the detailed version of the lecture slides.

4. Literature

- Tutorial by [Burgess](#) (1998)
- Regarding kernel methods: [Müller et al.](#) *An Introduction to Kernel-Based Learning Algorithms*, IEEE Trans. on Neural Networks (2001)
- Classical textbook, with background on SVMs and VC-dimension (not covered here), by [Vapnik](#) on *Statistical Learning Theory*, Wiley (1998)